

Unit 2: Cells, Development, Biodiversity and Conservation

IAS compulsory unit

Externally assessed

Unit description

Introduction

This unit starts with the cell as the basic unit of all living organisms, leading to cell division, formation of gametes, fertilisation and the continuity of life. The roles of stem cells, gene expression, and the influence of the environment and epigenetics on phenotypes are also included. Cell development leads to an understanding of the structure and functions of plant cells, and how plants may be exploited by humans for fibres and as sources of drugs. This unit also considers the diversity of life and how biodiversity can be measured. The unit ends with an account of reasons for changes in populations over time, and the methods used by zoos and seed banks for the conservation of endangered species and their genetic diversity.

Practical skills

In order to develop their practical skills, students should be encouraged to carry out a range of core practical experiments related to this topic. There are also opportunities to carry out additional practical work, such as investigating factors affecting the growth of pollen tubes.

Mathematical skills

There are opportunities for the development of mathematical skills in this unit, including using ratios, percentages and fractions, substituting values into algebraic equations, calculation of magnification, understanding the terms *mean*, *mode* and *median*, constructing and interpreting frequency tables, bar charts and histograms. (Please see *Appendix 6: Mathematical skills and exemplifications* for further information.)

Assessment information

- First assessment: June 2019.
 - The assessment is 1 hour and 30 minutes.
 - The assessment is out of 80 marks.
 - Candidates must answer all questions.
 - The paper may include multiple-choice, short open, open-response, calculations and extended-writing questions.
 - The paper will include a minimum of 8 marks that target mathematics at Level 2 or above.
 - Calculators may be used in the examination. Please see *Appendix 8: Use of calculators*.
 - Candidates will be expected to apply their knowledge and understanding to familiar and unfamiliar contexts.
-

Topic 3 – Cell Structure, Reproduction and Development

Candidates will be assessed on their ability to:

3.1	know that all living organisms are made of cells, sharing some common features
3.2	understand how the cells of multicellular organisms are organised into tissues, tissues into organs, and organs into organ systems
3.3	(i) know the ultrastructure of eukaryotic cells, including nucleus, nucleolus, ribosomes, rough and smooth endoplasmic reticulum, mitochondria, centrioles, lysosomes and Golgi apparatus (ii) understand the function of the organelles listed in (i)
3.4	understand the role of the rough endoplasmic reticulum (rER) and the Golgi apparatus in protein transport within cells, including their role in the formation of extracellular enzymes
3.5	(i) know the ultrastructure of prokaryotic cells, including cell wall, capsule, plasmid, flagellum, pili, ribosomes and circular DNA (ii) understand the function of the structures listed in (i)
3.6	be able to recognise the organelles in 3.3 from electron microscope (EM) images
3.7	(i) know how magnification and resolution can be achieved using light and electron microscopy (ii) understand the importance of staining specimens in microscopy
3.8	CORE PRACTICAL 5 (i) use a light microscope to make observations and labelled drawings of suitable animal cells (ii) use a graticule with a microscope to make measurements and understand the concept of scale
3.9	(i) know that a locus is the location of genes on a chromosome (ii) understand the linkage of genes on a chromosome
3.10	understand the role of meiosis in ensuring genetic variation through the production of non-identical gametes as a consequence of independent assortment of chromosomes in metaphase I and crossing over of alleles between chromatids in prophase I <i>Names of the stages of prophase are not required.</i>
3.11	understand how mammalian gametes are specialised for their functions (including the acrosome in sperm and the zona pellucida in the egg cell)
3.12	know the process of fertilisation in mammals, including the acrosome reaction, the cortical reaction and the fusion of nuclei
3.13	know the process of fertilisation in flowering plants, starting with the growth of a pollen tube and ending with the fusion of nuclei
RECOMMENDED ADDITIONAL PRACTICAL Investigate factors affecting the growth of pollen tubes.	

3.15	CORE PRACTICAL 6 Prepare and stain a root tip squash to observe the stages of mitosis.
3.16	be able to calculate mitotic indices
3.17	(i) understand what is meant by the terms <i>stem cell</i> , <i>pluripotent</i> and <i>totipotent</i> , <i>morula</i> and <i>blastocyst</i> (ii) be able to discuss the ways in which society uses scientific knowledge to make decisions about the use of stem cells in medical therapies
3.18	understand how cells become specialised through differential gene expression, producing active mRNA, leading to the synthesis of proteins which, in turn, control cell processes or determine cell structure in animals and plants
3.19	understand how one gene can give rise to more than one protein through post-transcriptional changes to messenger RNA (mRNA)
3.20	(i) understand how phenotype is the result of an interaction between genotype and the environment (ii) know how epigenetic modification, including DNA methylation and histone modification, can alter the activation of certain genes (iii) understand how epigenetic modifications can be passed on following cell division
3.21	understand how some phenotypes are affected by multiple alleles for the same gene, or by polygenic inheritance, as well as the environment, and how polygenic inheritance can give rise to phenotypes that show continuous variation